

LUCIDPREP

Grade 8 Physical Science

U.S. Standards-Aligned Course Syllabus

PHYSICAL SCIENCE

Physical Science · NGSS

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

There is no single national science curriculum, but the Next Generation Science Standards (NGSS) are the dominant U.S. reference. NGSS middle-school standards are written as a 6–8 band; this syllabus places Physical Science at Grade 8, a common arrangement, with Earth & Space connections. Districts using integrated sequences will redistribute these standards across grades 6–8.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Assessment is continuous and mastery-based. Each week pairs low-stakes practice (five warm-ups and two homework sets) with a 40-question weekly assessment. Students scoring below 70% review the instructional presentation and retake, so mastery is reached before progressing. Module assessments at Weeks 9, 18 and 27 and a cumulative final at Week 36 measure retention and synthesis.

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Grade 8 Physical Science aligned
to the Next Generation Science
Standards

This 36-week course is designed around **the NGSS Middle School Physical Science standards (MS-PS1-MS-PS4) with Earth & Space connections (MS-ESS1), Science & Engineering Practices and Crosscutting Concepts**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	Grade 8
Course	Physical Science · NGSS
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Model the structure and properties of matter and its interactions.
- Analyze forces and predict changes in motion.
- Explain energy, its transfer and conservation.
- Describe how waves transfer energy and information.
- Connect physical science to Earth and space systems.

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Module 1 — Matter & Its Interactions

Week	Topics	Students will be able to	Standards
1 — Properties of Matter	Physical properties, states, classifying matter	Describe and classify matter by properties	MS-PS1-2
2 — Atoms & Molecules	Particle model; atoms, molecules, elements	Model the particle nature of matter	MS-PS1-1
3 — States of Matter & Changes	Phase changes and thermal energy	Explain state changes with the particle model	MS-PS1-4
4 — Substances & Mixtures	Elements, compounds, mixtures, solutions	Distinguish pure substances and mixtures	MS-PS1-1
5 — Chemical Reactions	Evidence of reactions; new substances	Identify evidence of chemical reactions	MS-PS1-2
6 — Conservation of Mass	Atoms rearranged; mass conserved	Explain conservation of mass	MS-PS1-5
7 — Reactions & Energy	Exothermic and endothermic processes	Analyze energy change in reactions	MS-PS1-6
8 — Synthetic Materials	How materials come from natural resources	Evaluate synthetic materials from resources	MS-PS1-3
9 — Module 1 Assessment	Cumulative review of matter	Demonstrate mastery of Module 1	MS-PS1

Module delivery notes. Move from macroscopic observations to particle models. Use models, data and investigations throughout. NGSS middle-school standards are banded 6-8; districts arrange them differently, so pacing may shift.

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Module 2 — Forces & Motion

Week	Topics	Students will be able to	Standards
1 — Describing Motion	Position, speed, velocity, motion graphs	Describe and represent motion	MS-PS2
2 — Forces & Newton's Laws	Balanced/unbalanced forces and motion	Relate force to change in motion	MS-PS2-2
3 — Newton's Third Law	Action-reaction pairs; collisions	Apply Newton's third law	MS-PS2-1
4 — Engineering: Collision Forces	Designing to reduce collision impact	Design a solution to a collision problem	MS-PS2-1
5 — Gravity	Masses and gravitational attraction	Explain gravitational interactions	MS-PS2-4
6 — Electric & Magnetic Forces	Fields, attraction and repulsion	Analyze electric and magnetic forces	MS-PS2-3
7 — Fields & Non-Contact Forces	Field strength and distance	Investigate factors affecting forces	MS-PS2-5
8 — Forces in Systems	Applying force concepts to real systems	Apply force models to phenomena	MS-PS2
9 — Module 2 Assessment	Cumulative review of forces and motion	Demonstrate mastery of Module 2	MS-PS2

Module delivery notes. Move from macroscopic observations to particle models. Use models, data and investigations throughout. NGSS middle-school standards are banded 6-8; districts arrange them differently, so pacing may shift.

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Module 3 — Energy

Week	Topics	Students will be able to	Standards
1 — Kinetic Energy	Motion, mass and speed	Relate kinetic energy to mass and speed	MS-PS3-1
2 — Potential Energy	Stored energy from position/interaction	Model potential energy in a system	MS-PS3-2
3 — Energy Transfer	Transfer between objects and systems	Explain energy transfer	MS-PS3-5
4 — Thermal Energy & Heat	Temperature and thermal energy transfer	Analyze thermal energy transfer	MS-PS3-3
5 — Engineering: Thermal Device	Designing to control energy transfer	Design a device to control thermal energy	MS-PS3-3
6 — Energy & Temperature Change	Mass, type and temperature effects	Analyze factors affecting thermal energy	MS-PS3-4
7 — Energy Conservation	Transformations and tracking energy	Explain conservation of energy	MS-PS3-5
8 — Energy in Systems	Real-world energy analysis	Apply energy concepts to systems	MS-PS3
9 — Module 3 Assessment	Cumulative review of energy	Demonstrate mastery of Module 3	MS-PS3

Module delivery notes. Move from macroscopic observations to particle models. Use models, data and investigations throughout. NGSS middle-school standards are banded 6-8; districts arrange them differently, so pacing may shift.

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Module 4 — Waves & Earth/Space Systems

Week	Topics	Students will be able to	Standards
1 — Wave Properties	Amplitude, wavelength, frequency	Describe wave properties with a model	MS-PS4-1
2 — Wave Behavior	Reflection, absorption, transmission	Explain how waves interact with materials	MS-PS4-2
3 — Sound Waves	Vibrations, medium, pitch and loudness	Analyze sound as a wave	MS-PS4-1/2
4 — Light & the EM Spectrum	Light behavior and the spectrum	Analyze light and the EM spectrum	MS-PS4-2
5 — Digital Signals & Information	Analog vs digital; encoding information	Explain wave-based information transfer	MS-PS4-3
6 — The Solar System	Earth-Sun-Moon, gravity in space, planets	Model the solar system and gravity	MS-ESS1-2/3
7 — Earth, Moon & Sky	Seasons, moon phases, eclipses	Explain cyclic patterns from Earth-Sun-Moon	MS-ESS1-1
8 — Systems Connections	Connecting physical science to Earth/space	Apply physical science to Earth/space systems	MS-ESS1
9 — Final Assessment	Cumulative final across all modules	Demonstrate mastery of the full course	MS-PS1-PS4

Module delivery notes. Move from macroscopic observations to particle models. Use models, data and investigations throughout. NGSS middle-school standards are banded 6–8; districts arrange them differently, so pacing may shift.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	NGSS: MS-PS1, MS-PS2, MS-PS3, MS-PS4 (with MS-ESS1)
Science & Engineering Practices	Asking questions and defining problems; developing and using models; planning and carrying out investigations; analyzing and interpreting data; using mathematics and computational thinking; constructing explanations and designing solutions; engaging in argument from evidence; obtaining, evaluating and communicating information.
Crosscutting Concepts	Patterns; cause and effect; scale, proportion and quantity; systems and system models; energy and matter; structure and function; stability and change.
State mapping	NGSS adopted or adapted by nearly all states; middle-school standards are banded 6–8 and arranged differently by state.

Content Production Guidance

Author each week around an anchoring phenomenon or problem: 5 warm-ups, an explanatory lecture with models, 2 videos, an AI presentation with visual models and investigations, 2 homework sets, and a 40-question bank mixing three-dimensional items (SEP × DCI × CCC). Tag items to NGSS performance expectations.

Curriculum Basis & Source Notes

This Grade 8 science syllabus was generated from the Next Generation Science Standards. It reflects the Physical Science · NGSS course; because NGSS middle-school standards are banded 6–8, the arrangement here is one common domain-based model.

Primary reference	Next Generation Science Standards — MS (grades 6–8) performance expectations, with Science & Engineering Practices and Crosscutting Concepts.
Framework basis	A Framework for K-12 Science Education (National Research Council).
State cross-check	California and other NGSS-adopting state science frameworks.
LucidPrep alignment	Mapped to LucidPrep's 36-week, 4-module structure with 70% mastery.

Recommended next production step. Select anchoring phenomena per module and commission three-dimensional item banks (SEP × DCI × CCC) tagged to NGSS performance expectations.